

Experiment 02: Building Computer Network using Router.

Introduction

In modern communication systems, routers play a vital role in connecting multiple networks and directing data packets to their correct destinations. Cisco Packet Tracer is a widely used simulation tool that allows students to design, configure, and test networks in a virtual environment without physical hardware. In this lab, a network topology is created and configured using a router to enable communication between different devices and networks. This practical activity helps in understanding routing concepts, IP addressing, and verifying connectivity in a simulated network setup.

Objectives

The main objectives of this lab are:

- To understand the role and functionality of routers in a computer network.
- To design a basic network topology using Cisco Packet Tracer.
- To configure router interfaces with appropriate IP addressing.
- To establish communication between different network segments through proper routing configuration.
- To verify network connectivity using simulation tools like ping and traceroute commands.
- To enhance practical knowledge of networking concepts such as routing, IP addressing, and packet forwarding.

Procedures of the Experiment:

- **Opening Cisco Packet Tracer Software:**

Launch the application and open a new workspace.

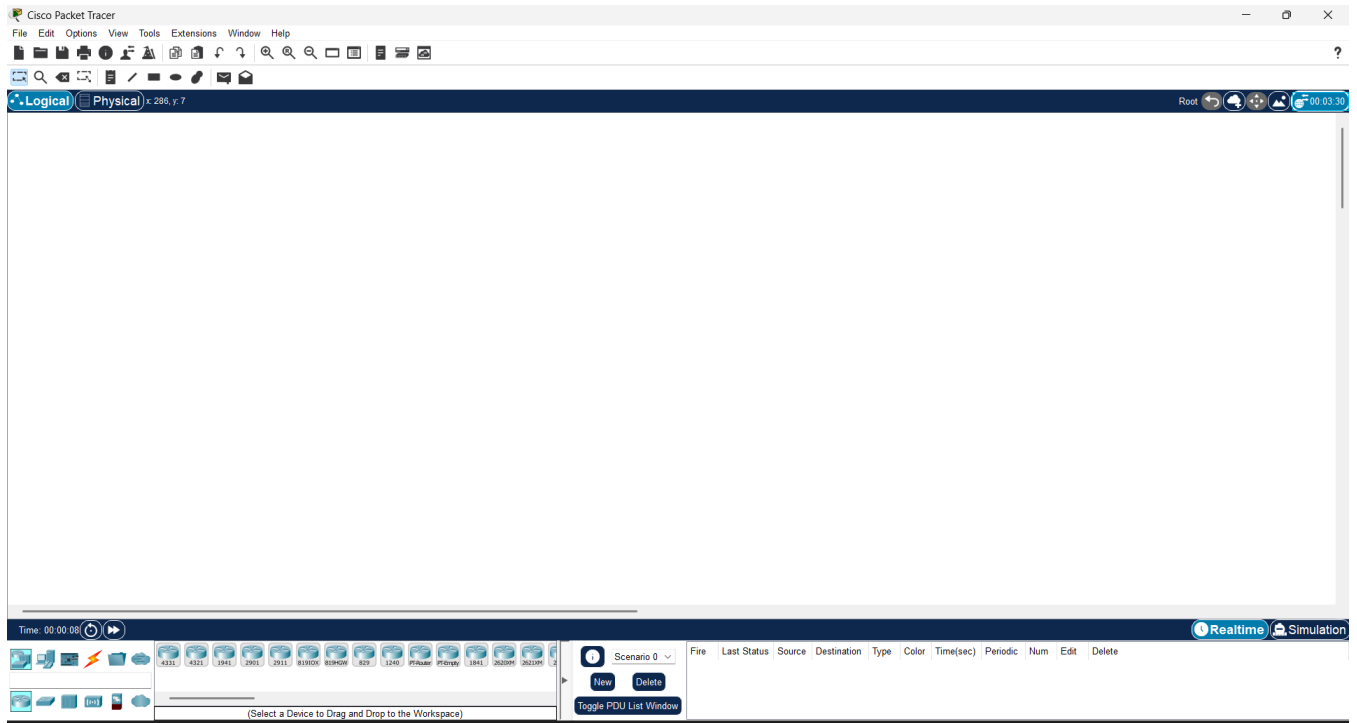


Figure:2.1

- **Adding Devices**

From The Devices Section of the Cisco Packet Tracer Software, drag and drop a router, two switches, and six PCs onto the workspace.

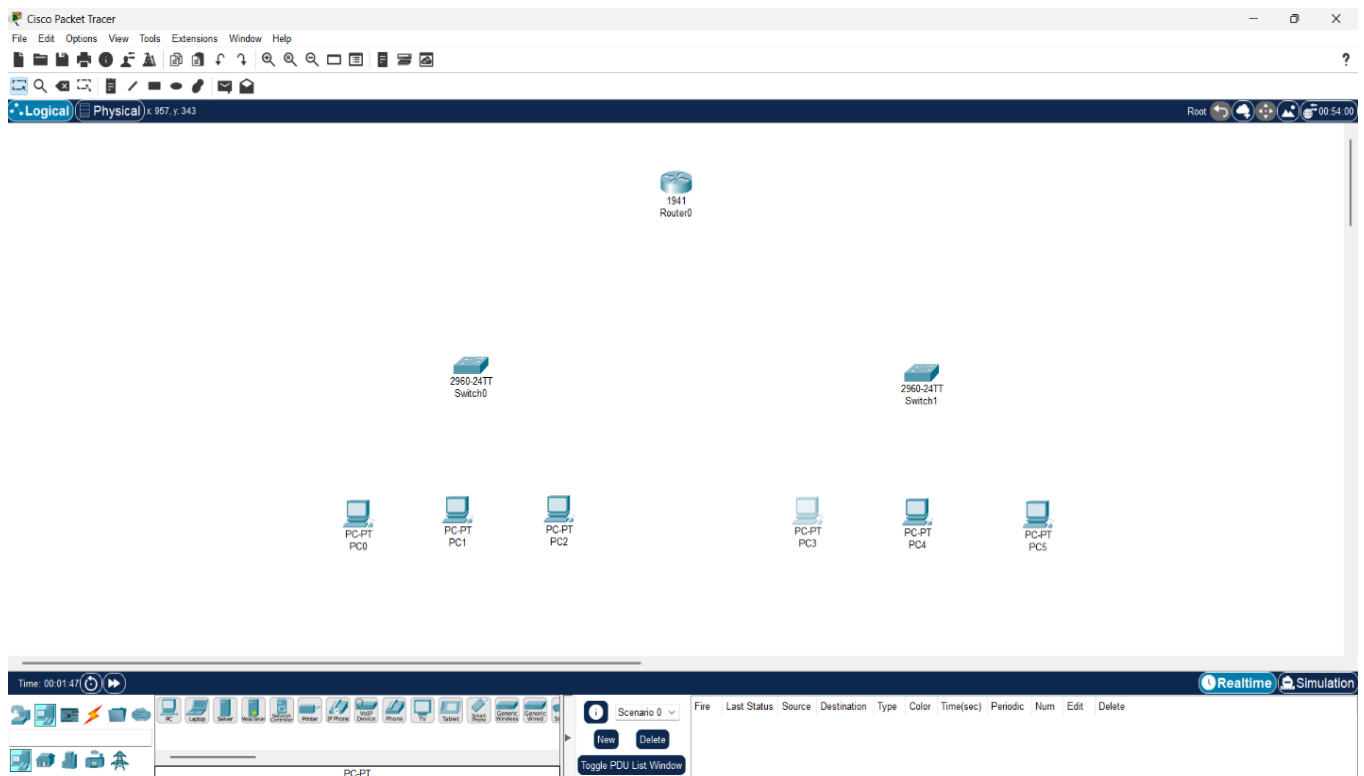


Figure:2.2

- **Connecting Devices**

- PC0, PC1 & PC2 → Switch0
- PC3, PC4 & PC5 → Switch1
- Switch0, Switch1 → Router1 (FastEthernet0/0, FastEthernet0/1)

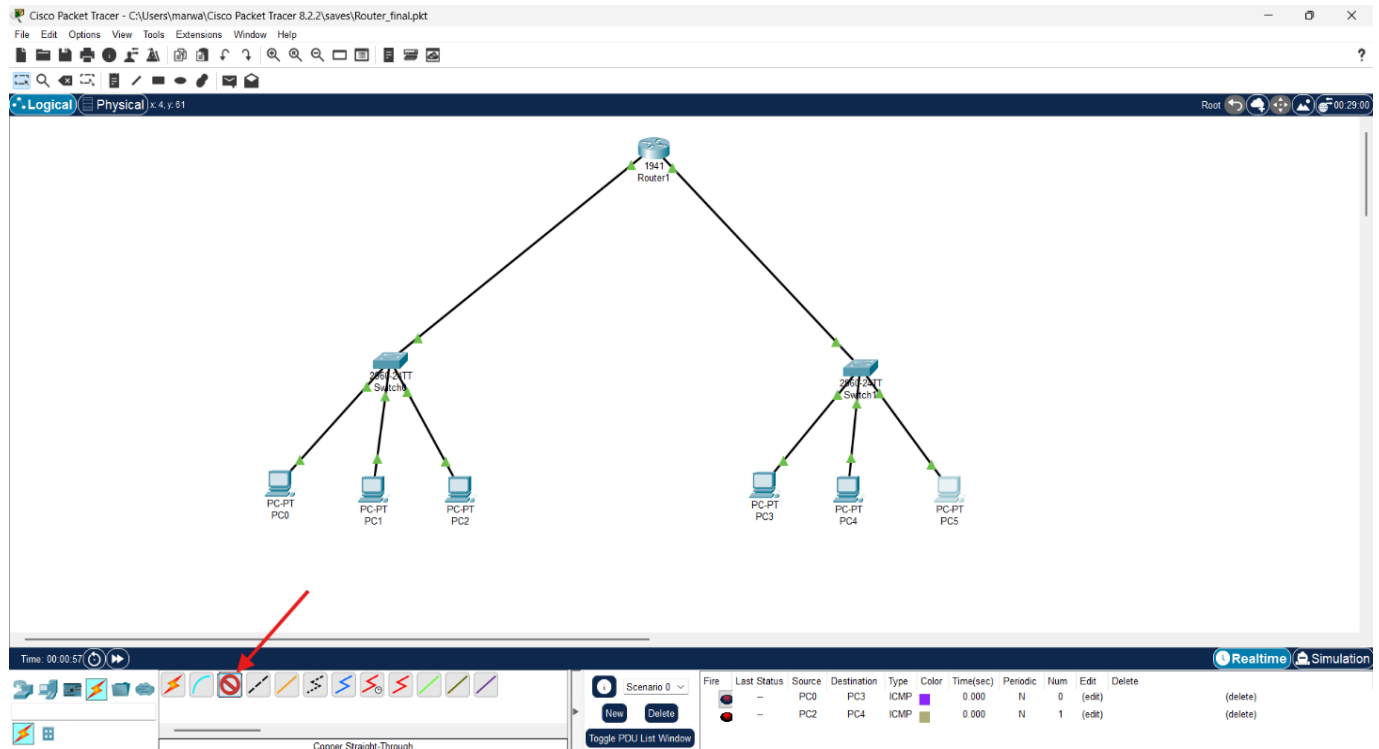


Figure:2.3

- **Assigning IP Address to Devices:**

- **Network 1:**

- PC0: 10.10.10.1 / 255.255.255.0
- PC1: 10.10.10.2 / 255.255.255.0
- PC2: 10.10.10.3 / 255.255.255.0

- **Network 2:**

- PC3: 192.168.0.1 / 255.255.255.0
- PC4: 192.168.0.2 / 255.255.255.0
- PC5: 192.168.0.3 / 255.255.255.0

- **Switch Interconnection:**

- Switch0: 10.10.10.4
- Switch1: 192.168.0.4

Add These Two IP addresses in the Router also in the Gateway Section of the corresponding Connected PC's.

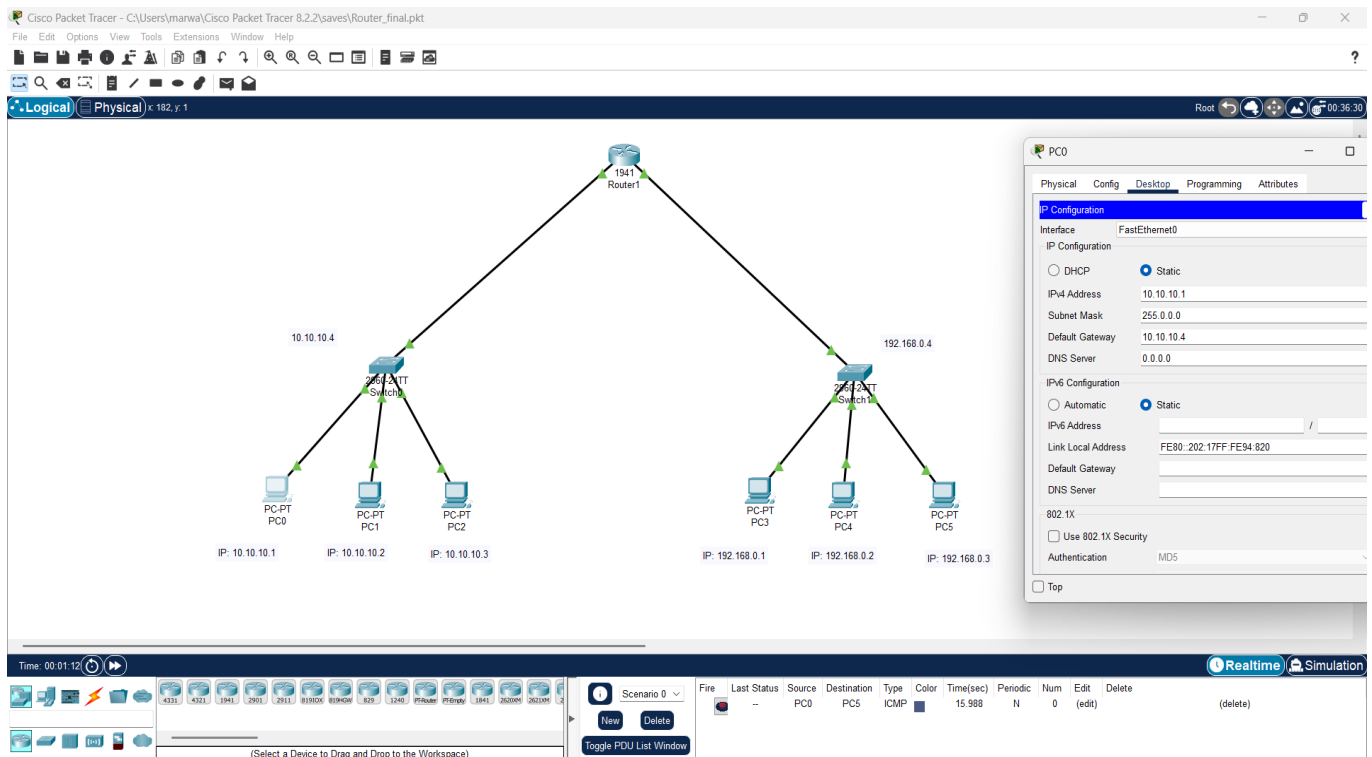


Figure:2.4

• Testing Connectivity

Test The Connectivity By sliding the ICMP packets from a pc connected with the first switch to a pc that is connected with the second Switch (for example: From the PC0→PC5 and then running the simulation. Here is the figure of the Successful Network setup.

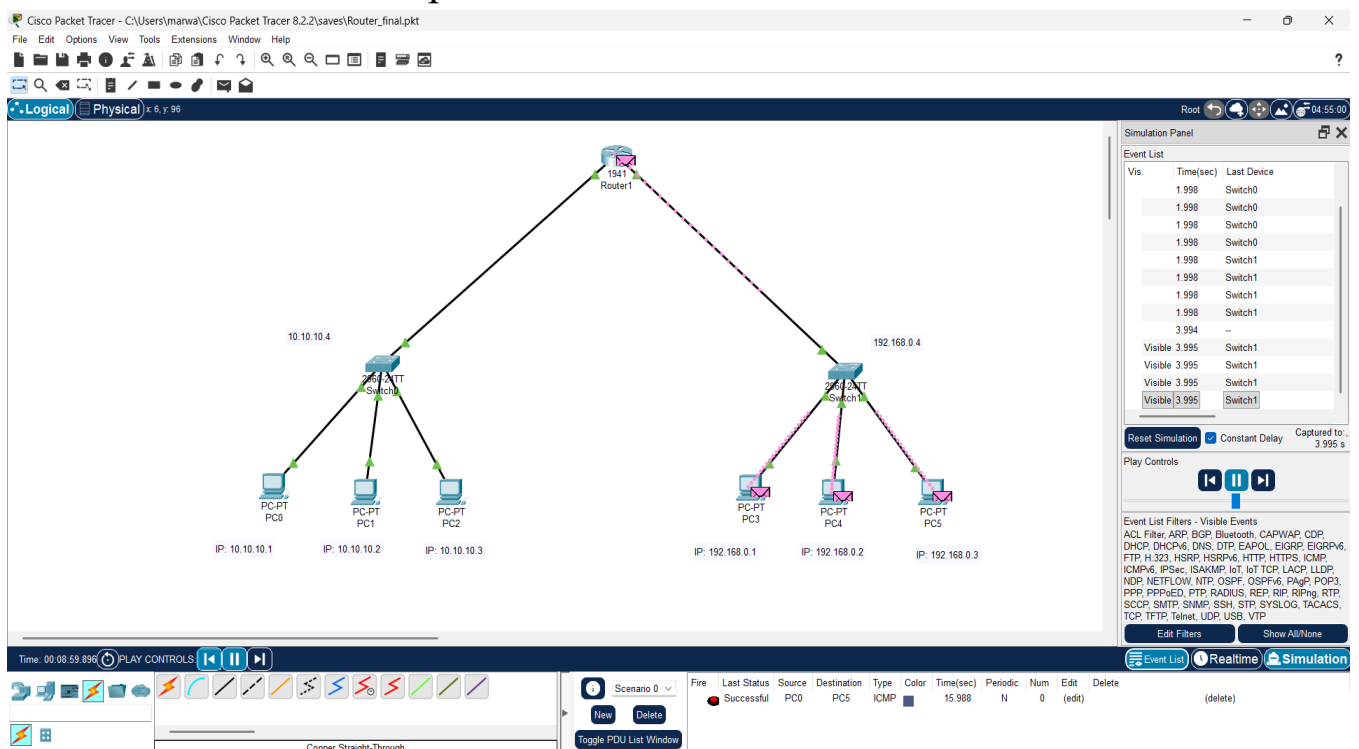


Figure:2.5

Result of the Experiment

All PCs from both networks communicated successfully through the routers. Static routing enabled data exchange between the two networks.

Conclusion

This Lab Report demonstrated how routers connect multiple networks and route packets between them. The experiment confirmed that router is one of the most essential networking device for inter-network communication.